

Mechanical Engineering Experiments — Report 4

機械工程實驗(一) 第四次報告

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Q1:

(a) .LabVIEW:

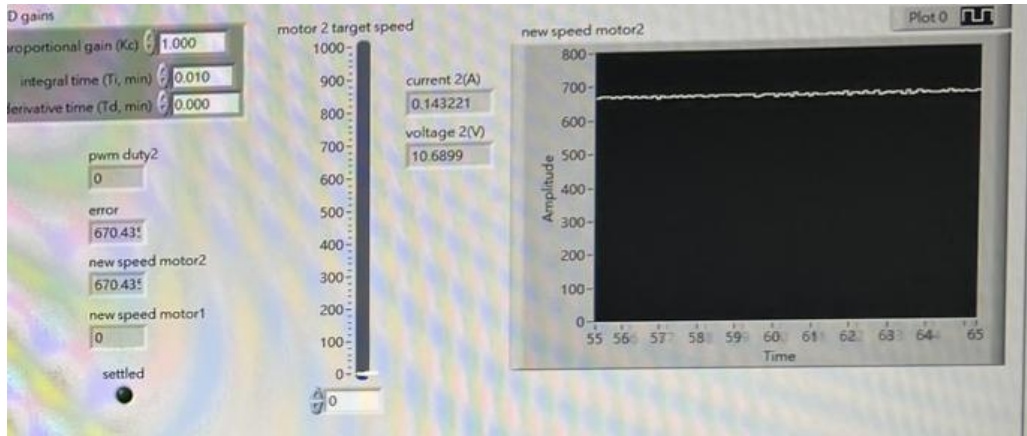


Figure1. Angular Speed from LabVIEW

$$\omega = 670.435 \text{ rpm} = 670.435 \times \frac{2\pi}{60} \text{ rad/s} \approx 70.208 \text{ rad/s}$$

$$I \approx 0.1432 \text{ A} , V = 10.6899 \text{ V}$$

By datasheet:

$$\text{Torque Constant, } k_t = 0.91 \frac{\text{kgf} \times \text{cm}}{\text{A}} = 0.91 \times 9.81 \div 100 \frac{\text{N} \times \text{m}}{\text{A}} \approx 0.0893 \frac{\text{N} \times \text{m}}{\text{A}}$$

$$\text{Voltage Constant, } k_e = 8.68 \frac{\text{mV}}{\text{r/min}} = 8.68 \times 0.001 \times \frac{60}{2\pi} \frac{\text{V}}{\text{rad/s}} \approx 0.0829 \frac{\text{V}}{\text{rad/s}}$$

$$\text{Coil Resistance, } \Omega_{\text{provided}} = 5\Omega$$

By measurement:

$$m = 0.2192 \text{ kg} , r = 0.1 \text{ m}$$

$$B = \frac{k_t i(t)}{\omega_m(t)} = \frac{0.0893 \times 0.1432}{70.208} \approx 1.821 \times 10^{-4} \frac{\text{N} \times \text{m}}{\text{rad/s}}$$

$$J = 0.5mr^2 = 0.5 \times 0.2192 \times 0.001 \approx 1.1 \times 10^{-4} \text{ kg} \times \text{m}^2$$

$$\Omega_{\text{measured}} = 9\Omega , L = 1.390 \text{ mH}$$

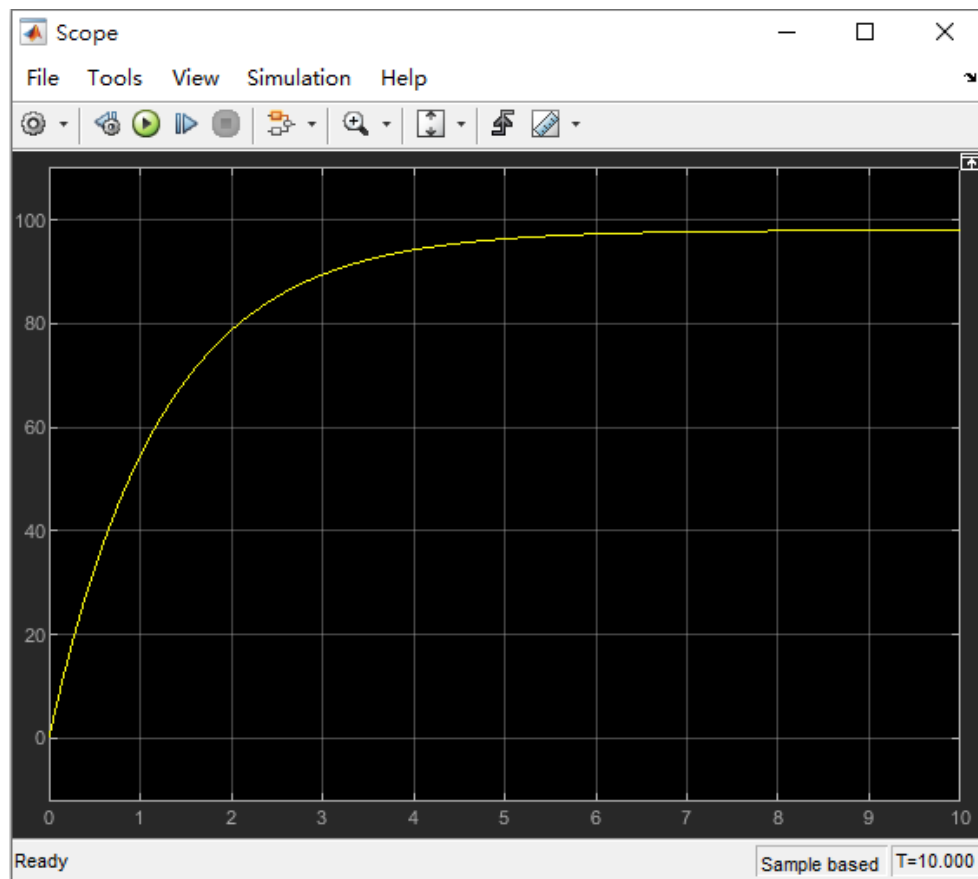


Figure2. Simulated Angular Speed from Measured data

$$\omega_{simulate,measured} \approx 98 \text{ rad/s}$$

$$steady \text{ error} = \frac{98-70.208}{70.208} \approx 39.585\%$$

By provided data:

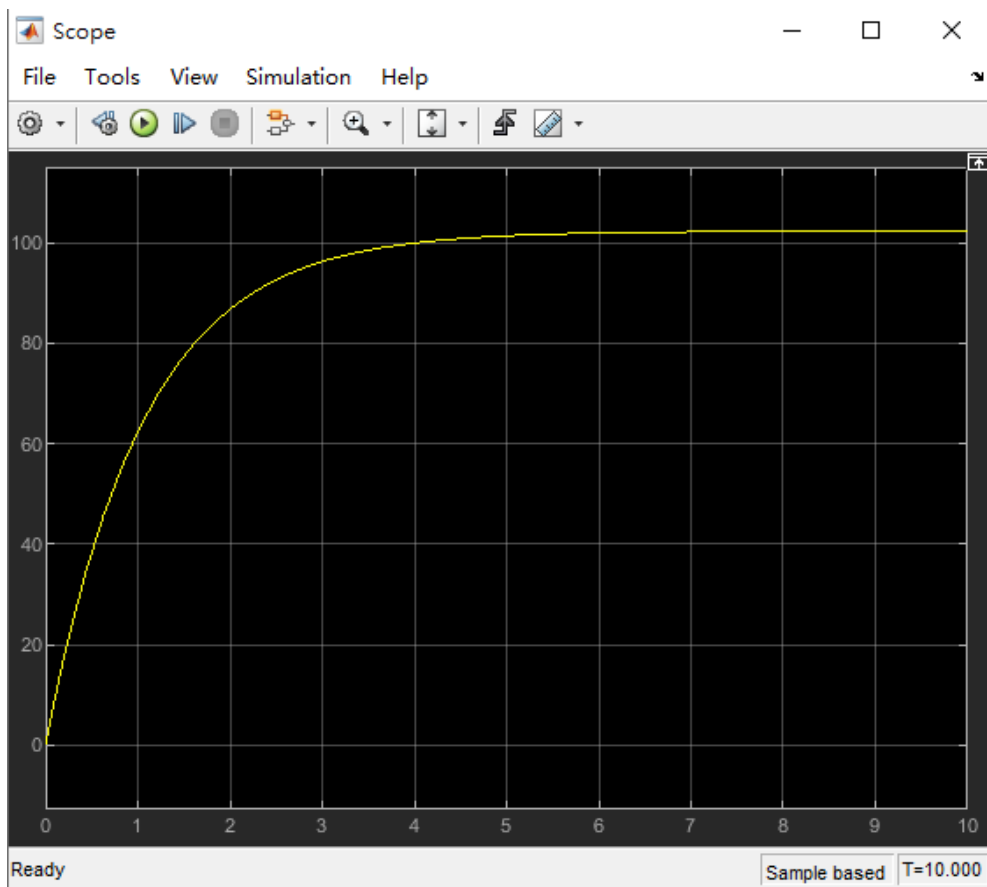


Figure3. Simulated Angular Speed from Provided data

$$\omega_{simulate,provided} \approx 102 \text{ rad/s}$$

$$steady \text{ error} = \frac{102-70.208}{70.208} \approx 45.283\%$$

From two steady errors we calculated, it seems that the measured data shows less steady errors from the reference value, which provides a better result. However, we can see that both two data have a large error. We speculate the possible reason is the measuring inaccuracy of resistance and inductance from the motor wire.

(b) .

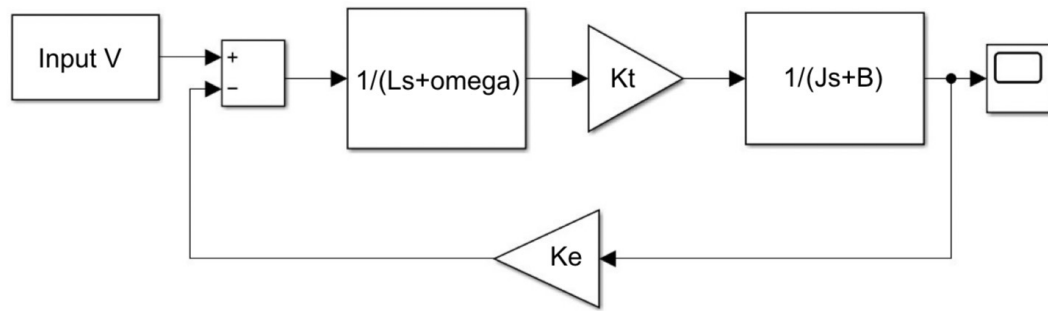


Figure4. Block Diagram of motor

$$G_e(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{k_t}{k_t k_e + (Ls + \Omega)(Js + B)}$$

$$Y(s) = \omega(s) = G_e(s)R(s) = G_e(s)V(s)$$

$$V(s) = \frac{10.6899}{s}, \quad k_t = 0.0893 \frac{N \times m}{A}, \quad k_e = 0.0829 \frac{V}{rad/s}$$

$$\Omega_{measured} = 9\Omega, \quad B = 1.821 \times 10^{-4} \frac{N \times m}{rad/s}$$

$$\lim_{s \rightarrow 0} sY(s) = s \times \frac{10.6899}{s} \times \frac{k_t}{LJs^2 + (\Omega J + BL)s + (k_t k_e + \Omega B)} = 10.6899 \times$$

$$\frac{0.0893}{0.0893 \times 0.0829 + 9 \times 1.821 \times 10^{-4}} = 105.576 \text{ rad/s}$$

$$\text{steady state error} = \lim_{s \rightarrow 0} s(R(s) - Y(s)) = 70.208 - 105.576 =$$

$$\mathbf{-35.368 \text{ rad/s}}$$

Q2:

(a) .

For 200 RPM:

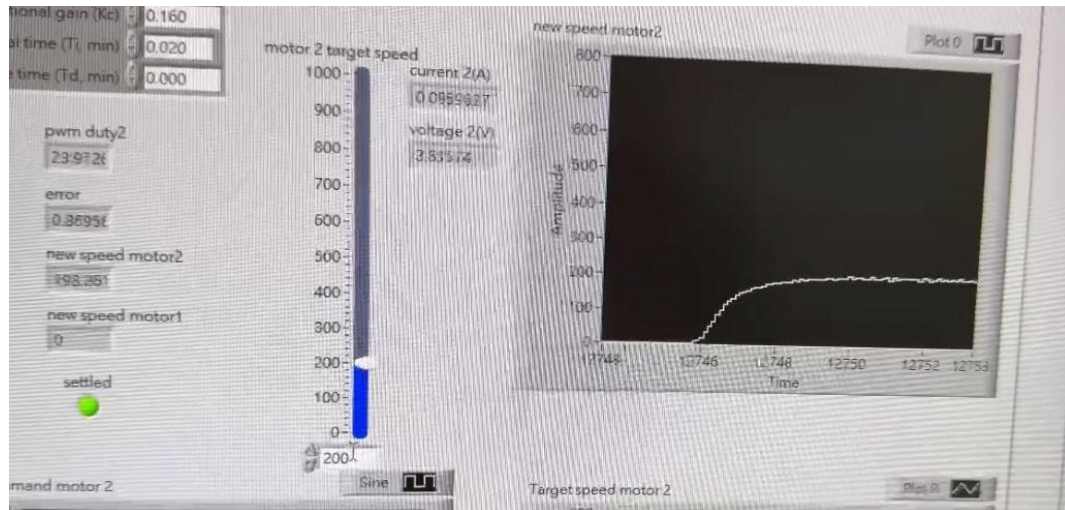


Figure5 . Image of Target Speed of 200 RPM from LabVIEW

($K_c = 0.16$, $T_i = 0.02$)

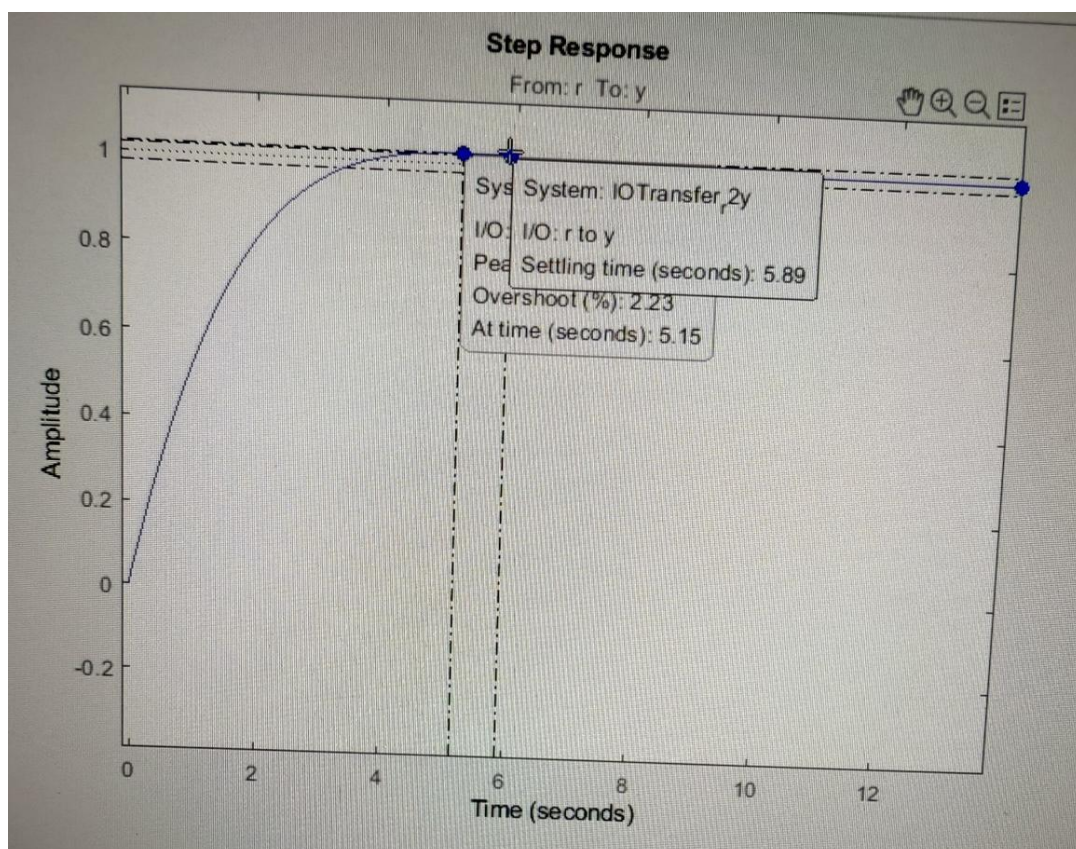


Figure6. Image of Target Speed of 200 RPM from Simulink

($K_c = 0.16$, $T_i = 0.02$)

For 400 RPM:

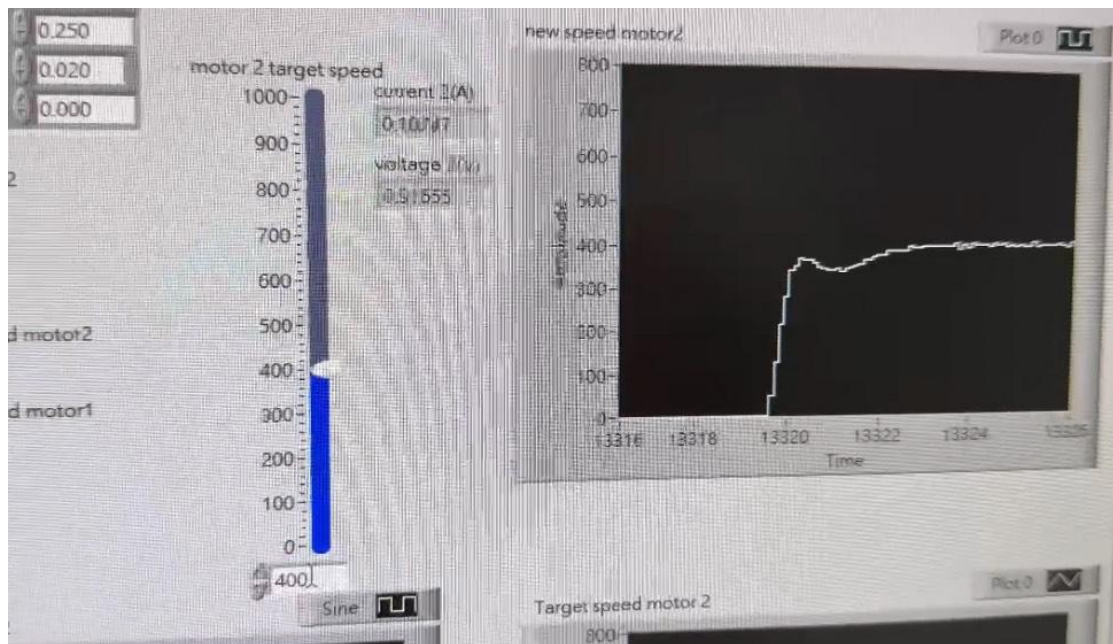


Figure7 . Image of Target Speed of 400 RPM from LabVIEW

($K_c = 0.25$, $T_i = 0.02$)

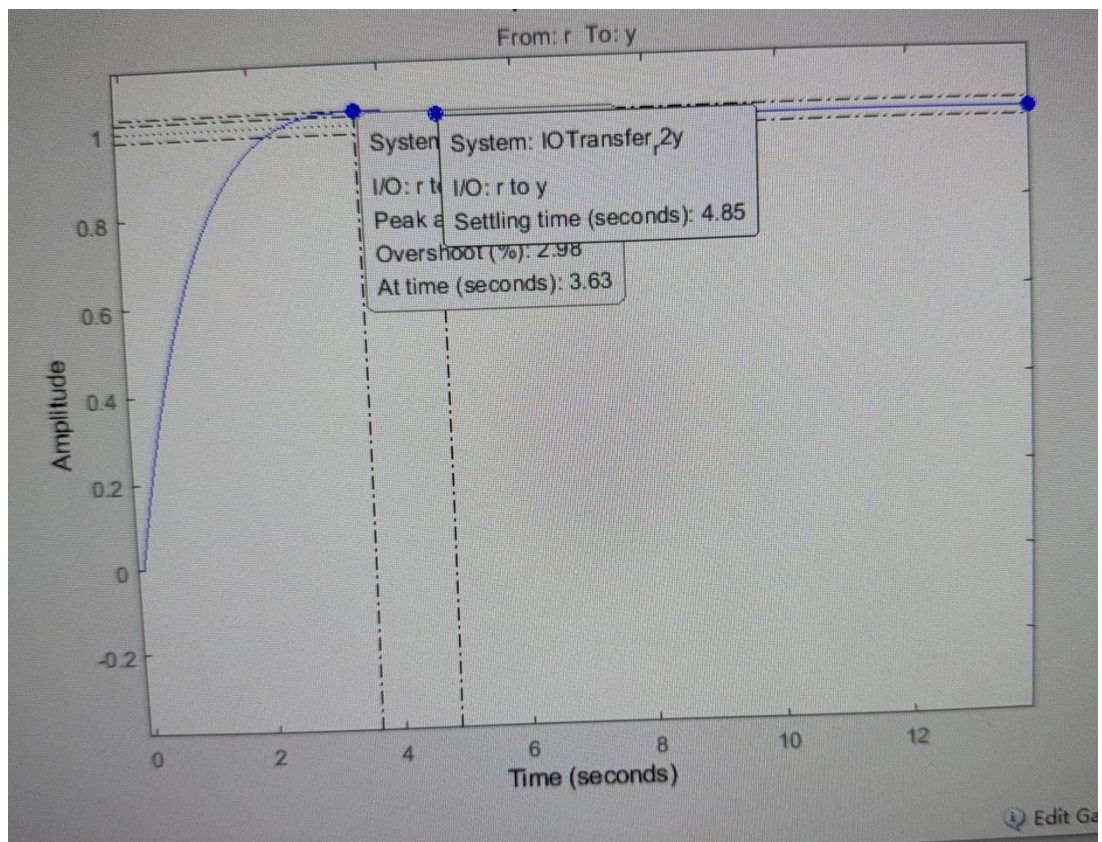


Figure8. Image of Target Speed of 400 RPM from Simulink

($K_c = 0.25$, $T_i = 0.02$)

For 600 RPM:

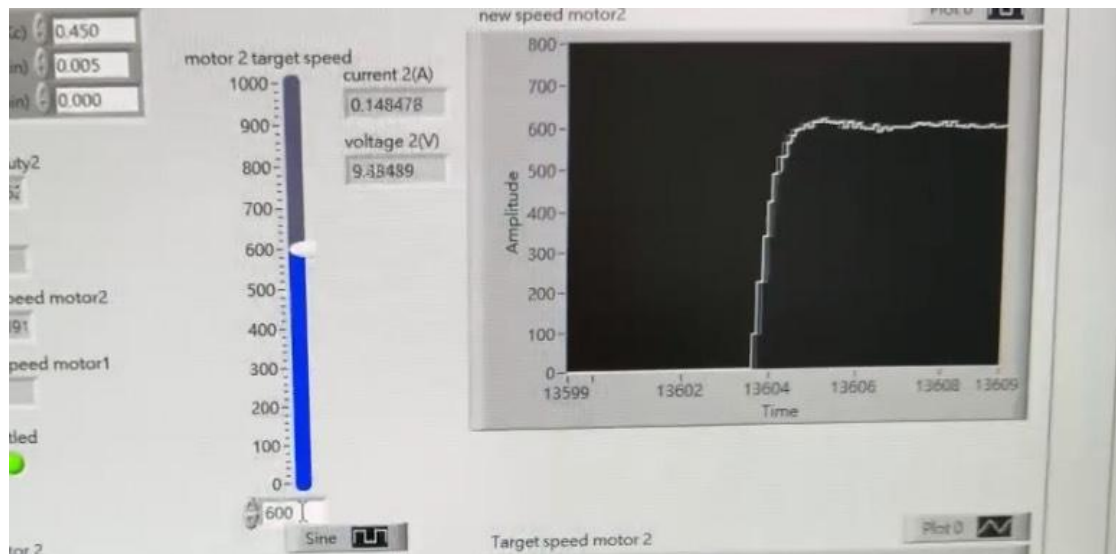


Figure9 . Image of Target Speed of 600 RPM from LabVIEW

($K_c = 0.45$, $T_i = 0.005$)

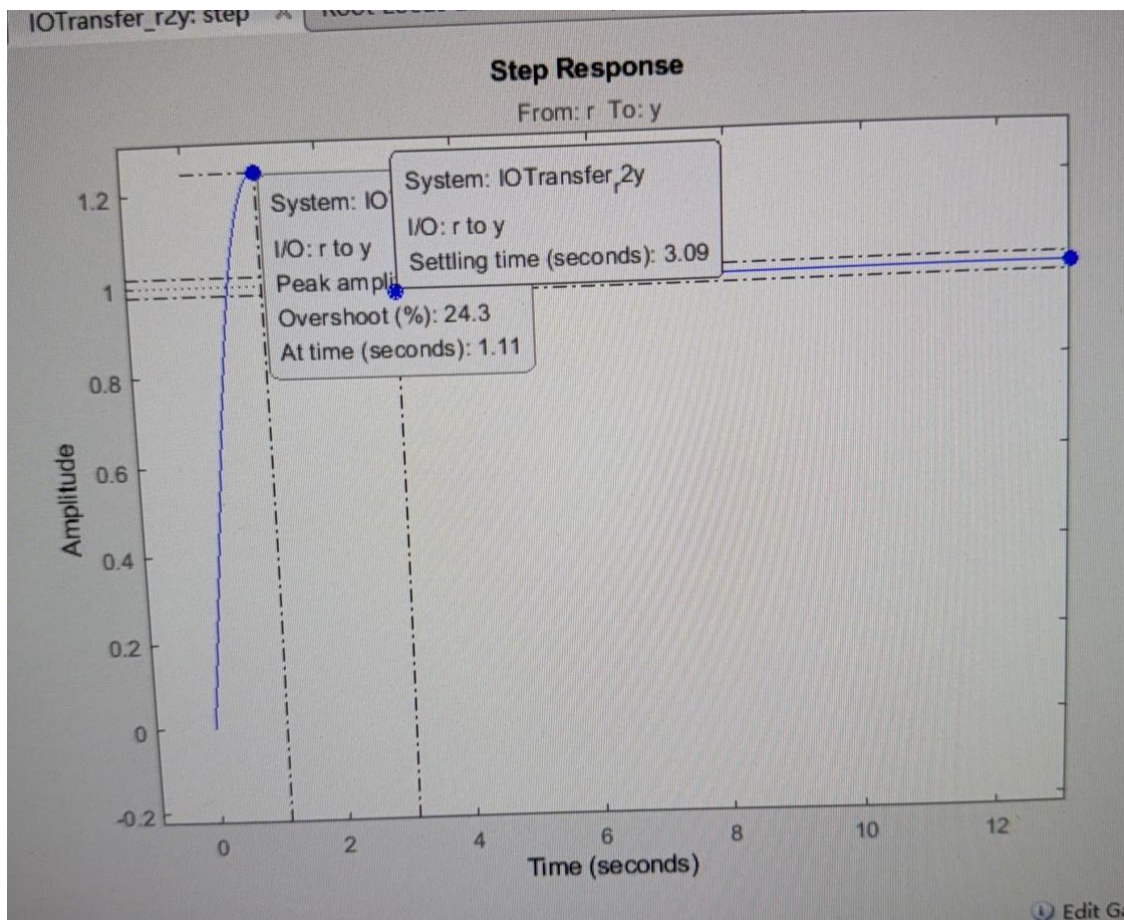


Figure10. Image of Target Speed of 600 RPM from Simulink

($K_c = 0.45$, $T_i = 0.005$)

(b) . If we fix one coefficient while tuning the other one, we can easily observe the result difference by the one we are tuning. For example, if we want to know how P control, K_c , affects our response of the motor, we can fix the value of I control, T_i , and change the K_c value. By comparing the result with the unchanged one, we can know how P control affects the response.

P Control:

Determines how aggressively the motor generates torque/current in response to instantaneous error.

If we increase the K_c value, the overshoot percentage will increase while having a minimal impact on settling time.

I control:

It accumulates error over time and keeps increasing torque until steady state error ≈ 0 .

If we increase the T_i value, both overshoot percentage and settling time will increase.